Stem cells

☞ **1** Replace the missing vowels (a, e, i, o, u) in the first two paragraphs.

A single f__rt__l__s__d egg divides into __v__ry d__ff__r__nt type of c__ll that exists in an adult. V__ry early in the lif__ of the div__ding egg there are **stem cells**, which can d__v__d__ to change into specialist cells such as m__scle cells or blo__d cells.

There are two types of stem cells – **embryo** stem cells and **adult** stem cells. Embryo st__m c__lls can gr__w into any type of c__ll. Ad__lt st__m cells __nly become the type of cell of the t__ss__ __ they live in. Some adult stem c__lls can be made to b__h__ve like __mbry__ cells to pr__d__ce any type of cell. These are c__ll__d 'induced pluripotent' st__m c__lls.

Scientists hope to use stem cells to grow new cells in damaged tissue. This could help to cure diseases. For example, stem cells could be used to grow new pancreas cells in diabetics, heart cells in heart attack sufferers, and so on. High priority will be given to treating spinal injuries and disorders where cells stop working properly. Doctors will need to ensure that injected stem cells are not rejected by the body, which could see the new cells as invaders, and that the stem cells don't cause new problems.

☞ **2** What do fertilised egg cells divide into? _____

3 Muscle and blood cells are examples of what types of cells?

_____

4 What can adult 'induced pluripotent' stem cells do?

_____

5 What is it that scientists hope will grow new cells in damaged tissue?

_____

6 Where would a diabetic have stem cells implanted? _____

7 Why do you think spinal injuries will be a high priority? _____

_____

8 What are two of the problems that doctors have to solve with stem cell treatments?

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